

MITHUN KV

+91 6361916073 | kvmithun1234@gmail.com | Bangalore, Karnataka | LinkedIn | GitHub

SUMMARY

AI/ML engineer with hands-on experience building machine learning, Generative AI, and RAG applications using Python. Experienced in developing ML pipelines, LLM-powered systems, backend APIs, and deploying applications using Docker, AWS, and CI/CD.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **AI / Machine Learning:** Scikit-learn, TensorFlow, Deep Learning, Transformers, NLP
- **Generative AI / LLM:** LLM APIs, Prompt Engineering, Structured Outputs, Pydantic, RAG, RAG Evaluation, Embeddings, Qdrant, Semantic Search, Agents, Tool Use, LangGraph, MCP, Multi-Agent Systems, Observability, Guardrails, LLM Security, Fine-Tuning, Deployment
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn, Statistics
- **Backend / APIs:** FastAPI, Flask
- **MLOps / Cloud:** Docker, AWS S3, AWS ECR, AWS EC2, GitHub Actions, CI/CD, DVC, MLflow
- **Databases:** MongoDB, MySQL
- **Developer Tools:** Git, GitHub, Jupyter, VS Code, PyCharm
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

InsureFlow — End-to-End MLOps Insurance Prediction System

GitHub

Python, Scikit-learn, MongoDB, FastAPI, Docker, AWS, GitHub Actions

- Engineered an end-to-end vehicle-insurance cross-sell ML pipeline covering MongoDB ingestion, schema-based validation, feature engineering, and feature transformation with scaling.
- Applied SMOTEENN to handle class imbalance and trained a Random Forest classifier, evaluating performance with accuracy, precision, recall, and F1 metrics.
- Containerized the FastAPI prediction service with Docker, deployed it via Amazon ECR and EC2 with model artifacts stored in Amazon S3, and automated releases through GitHub Actions with a self-hosted EC2 runner for CI/CD.

YouTube Reviser — RAG-Based Video Revision Assistant

GitHub

Python, yt-dlp, Argos Translate, Sentence Transformers, Qdrant, Groq

- Built a RAG-based revision assistant that extracts transcripts from 25 Hindi educational YouTube videos, translates them to English, and chunks them with timestamp preservation.
- Generated 384-dimensional Sentence Transformer embeddings and indexed roughly 1,471 transcript chunks with meta-data in Qdrant for cosine-similarity semantic search.
- Designed a retrieval-to-generation pipeline using Qdrant and a Groq-hosted LLM (openai/gpt-oss-120b) to produce grounded answers with timestamp-aware source attribution.

Jan Suvidha — AI Welfare Assistance Platform

Django, Flask, MongoDB, Gemini API

- Built an AI-powered welfare assistance platform that collects citizen information and provides scheme/recommendation support based on eligibility.
- Integrated Gemini API through a separate AI service for natural-language assistance and AI-powered processing.
- Developed the Django-based citizen/admin application with MongoDB-backed data handling and supporting services for recommendations, reminders and SMS.

AI Resume Parser & Candidate Matching System

GitHub

Python, Groq API, Pydantic, PyPDF, python-docx, Flask

- Built an LLM-powered parser that extracts structured candidate information from PDF and DOCX resumes and converts job descriptions into structured data.
- Implemented Pydantic schema validation for reliable structured LLM outputs across the parsing pipeline.
- Developed candidate matching and ranking against job requirements and exposed the system through a Flask API.

EDUCATION

Presidency University

Bachelor of Technology in Computer Science Engineering (Data Science)

Aug 2023 – Aug 2027

CGPA: 7.5 / 10

ACHIEVEMENTS

- 3rd Place — Tech Biz Hackathon
- Selected for Dayananda Sagar DevHack 2.0 National Level Hackathon